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# Blockchain Lab
# EXP 10

**Aim**: Explore the Cryptocurrency Landscape

**Theory:**

This experiment explores the cryptocurrency landscape by combining real-world market data analysis with blockchain technology research. The focus asset for this analysis is **Chainlink (LINK)**, and data was collected from three reputable sources as per the lab requirement:

- **Financial Data (APIs)** – Services like **CoinGecko** and **CoinMarketCap** provided structured data on market capitalization, circulating supply, and historical 24-hour trading volumes, which are essential for calculating liquidity and asset dominance.
- **News & Social Platforms** – Platforms such as **CryptoPanic** (news aggregator) and **LunarCrush** (social sentiment) were used to track how public perception and breaking news correlate with price volatility.
- **Regulatory & Institutional Portals** – Websites like **The Block Pro** or official government portals (e.g., SEC, EU MiCA updates) provided data on legal frameworks and institutional entry.

**Cryptocurrency Landscape**

- **Definition** – Cryptocurrencies are decentralized digital assets built on blockchain networks, enabling secure, transparent, and immutable transactions without central authority.
- **Market Diversity** – Thousands of cryptocurrencies exist, each serving unique roles:
  - **Store of Value (SoV)** – Bitcoin (BTC), characterized by digital scarcity.
  - **Utility Tokens** – Chainlink (LINK), Ether (ETH), used to pay for network services.
  - **Stablecoins** – USDT, USDC, pegged to fiat (USD) to provide price stability.
  - **Governance Tokens** – UNI, AAVE, granting holders voting rights over protocols.
- **Volatility** – Prices are highly sensitive to market sentiment, technological developments, and "macro" regulatory drivers.
- **Data Analysis Importance** – Integrating financial, social, and regulatory data is essential for a holistic understanding of asset performance and market health.

**Data Collection Process**

The program used the CoinGecko API to automate historical data fetching for Chainlink. The retrieved data was processed with Pandas to compute:

- **Daily Returns** – Percentage change between consecutive days to measure volatility.
- **Moving Averages (MA7, MA30)** – Short-term and long-term trend indicators.

- **Cumulative Returns** – Growth factor over time.
- **Rolling Volatility** – Short-term risk measurement based on daily return fluctuations.

**Visualization & Dashboard**

Using Matplotlib, a six-panel dashboard was created to display:

- **Price Trend** – Daily price movement over the selected period.
- **Price vs Volume** – Correlation between trading activity and price changes.
- **Daily Returns Distribution** – Statistical spread of daily gains/losses.
- **Moving Averages** – Technical analysis trend lines for price movement.
- **Cumulative Returns** – Overall growth performance.
- **Rolling Volatility** – Market risk trends over time.

**Technological Analysis**

Beyond market data, the study covered advancements in blockchain technology, including:

**Consensus Mechanisms** – The shift toward Proof of Stake (PoS) for sustainability and Proof of History (PoH) for faster processing.

- **Scalability Solutions** – Implementation of Layer-2 Rollups (Optimistic and ZK-Rollups) and Sharding to solve the "Blockchain Trilemma."
- **Interoperability Protocols** – Tools like Chainlink CCIP and Polkadot's Relay Chain enabling seamless cross-chain data and value exchange.
- **Privacy Features** – Zero-Knowledge Proofs (ZKP) allowing verifiable transactions without revealing sensitive underlying data.
- **Technical Roadmaps** – Ethereum's focus on "The Surge" (100k+ TPS) and Solana's "Firedancer" client for institutional-grade reliability.

**Market Trends & Adoption**

- **Merchant Acceptance** – Significant surge in interest with major processors like Stripe and Shopify integrating stablecoins for global settlement.
- **Wallet Growth** – Increasing downloads of non-custodial wallets (MetaMask, Phantom), indicating a shift toward direct on-chain interaction.
- **Network Activity** – Analysis of transaction volumes vs. active addresses to distinguish between "whale" activity and organic retail growth.
- **Integration with Traditional Finance** – The rise of Real-World Asset (RWA) Tokenization, bringing traditional assets like bonds and real estate on-chain.
- **Regulatory Developments** – The implementation of frameworks like the EU's MiCA and U.S. FIT21 Act, providing a clear licensing roadmap for the industry.

**Utilizing Api for analysis and visualization : Code:**

```
import requests import
matplotlib.pyplot as plt import
pandas as pd

import datetime
```

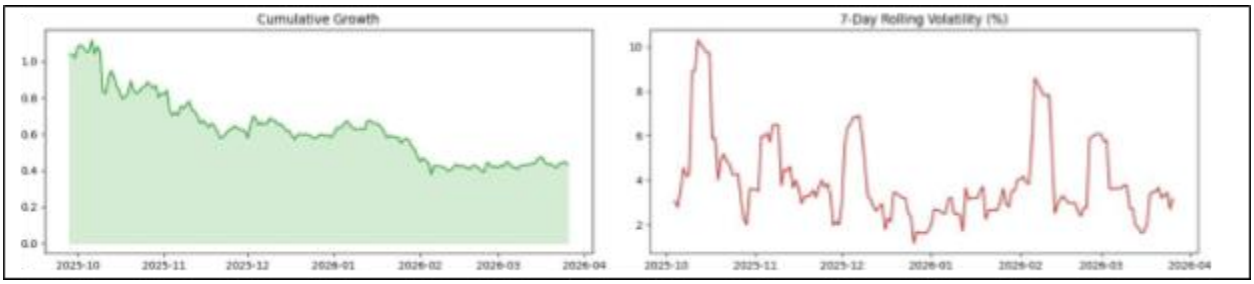
```
# ---------- CONFIGURATION---------- # Replace 'YOUR_API_KEY' with
the key you got from CryptoCompare
API_KEY = 'YOUR_API_KEY' SYMBOL = 'LINK' COMPARISON_SYMBOL =
'USD'
LIMIT = 180 # Number of days

# ---------- Fetch Data Function ---------- def
fetch_cryptocompare_data(symbol, comparison_symbol, limit,
api_key): url = f"https://min-
api.cryptocompare.com/data/v2/histoday" params = {
        "fsym": symbol, "tsym":
        comparison_symbol, "limit":
        limit,
        "api_key": api_key
    }
 response = requests.get(url,
    params=params) data = response.json()
 if data['Response'] == 'Success':
        # CryptoCompare returns OHLCV data in 'Data' ->
        'Data' return data['Data']['Data'] else:
        print(f"Error fetching data:
        {data.get('Message')}") return None

# ---------- Dashboard Function ---------- def
cryptocompare_dashboard(symbol, comparison_symbol, limit,
api_key): raw_data = fetch_cryptocompare_data(symbol,
comparison_symbol, limit, api_key) if not raw_data: return

    # Create DataFrame df =
    pd.DataFrame(raw_data)

    # Convert 'time' (unix timestamp) to datetime
    df['timestamp'] = pd.to_datetime(df['time'],
    unit='s') df.set_index('timestamp',
    inplace=True)

    # Standardize column names to match your
    previous logic # 'close' is used as the
    daily price df['price'] = df['close']
```

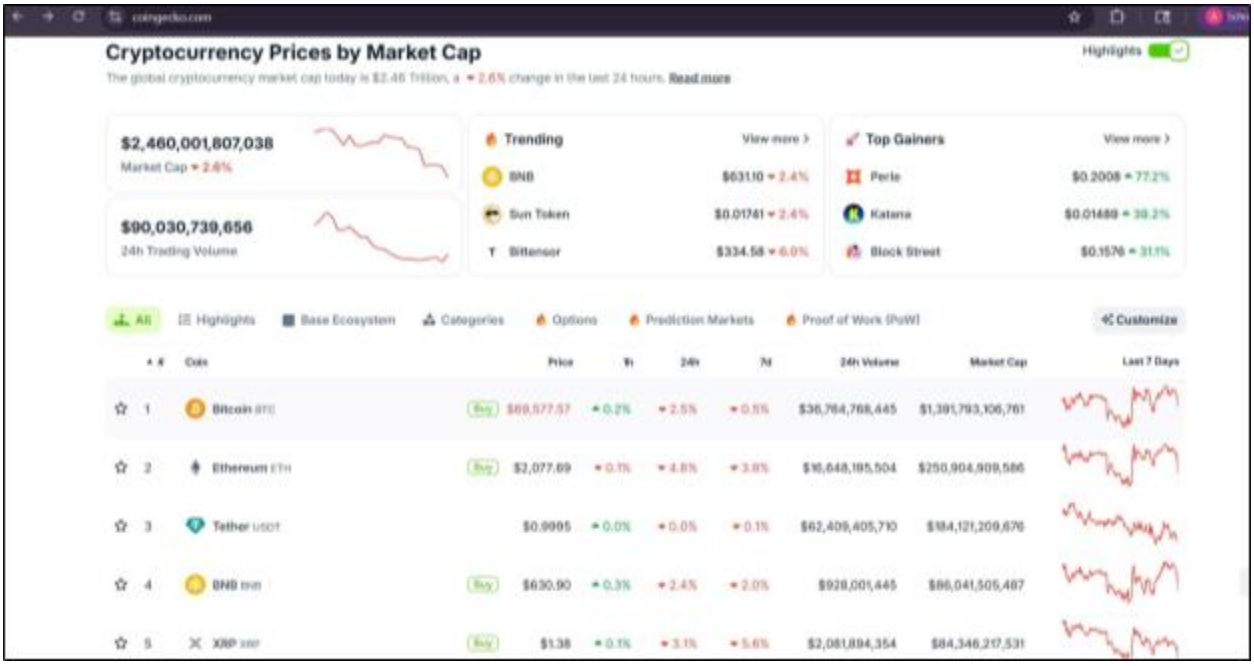
```
    df['volume'] = df['volumeto'] # Total volume in USD
    (volumeto)

    # Calculate metrics df['returns'] =
    df['price'].pct_change() df['MA7'] =
    df['price'].rolling(window=7).mean() df['MA30'] =
    df['price'].rolling(window=30).mean()
    df['cumulative'] = (1 + df['returns']).cumprod()
    df['volatility'] =
    df['returns'].rolling(window=7).std() * 100

    # ---------- Plotting----------
    fig, axes = plt.subplots(3, 2, figsize=(16, 12))
fig.suptitle(f'CryptoCompare: {symbol} Analysis Dashboard', fontsize=16,
fontweight='bold')

    # 1. Price Trend axes[0, 0].plot(df.index, df['price'],
    color='tab:blue', linewidth=2) axes[0,
    0].set_title('Historical Price Trend (USD)') axes[0,
    0].grid(True, linestyle='--', alpha=0.7)

    # 2. Price vs Volume axes[0, 1].plot(df.index,
df['price'], color='tab:blue', label='Price') axes[0,
1].bar(df.index, df['volume'], color='gray', alpha=0.3,
label='Volume (USD)') axes[0, 1].set_title('Price vs
Trading Volume') axes[0, 1].legend()

    # 3. Daily Returns Distribution axes[1,
    0].hist(df['returns'].dropna(), bins=40,
    color='tab:purple', edgecolor='white') axes[1,
    0].set_title('Daily Returns Distribution')

    # 4. Moving Averages axes[1, 1].plot(df.index,
df['price'], color='lightgray', alpha=0.5, label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='tab:orange',
label='7-Day MA') axes[1, 1].plot(df.index, df['MA30'],
color='tab:red', label='30-Day MA') axes[1,
1].set_title('Trend Analysis (Moving Averages)') axes[1,
1].legend()
```

```
    # 5. Cumulative Returns axes[2,
0].fill_between(df.index, df['cumulative'],
color='tab:green', alpha=0.2) axes[2,
0].plot(df.index, df['cumulative'], color='tab:green')
axes[2, 0].set_title('Cumulative Growth')

    # 6. Volatility axes[2, 1].plot(df.index,
    df['volatility'], color='tab:red') axes[2,
    1].set_title('7-Day Rolling Volatility (%)')
 plt.tight_layout(rect=[0, 0.03, 1,
    0.95]) plt.show()

# ---------- Run ----------
cryptocompare_dashboard(SYMBOL, COMPARISON_SYMBOL, LIMIT,
API_KEY)
```

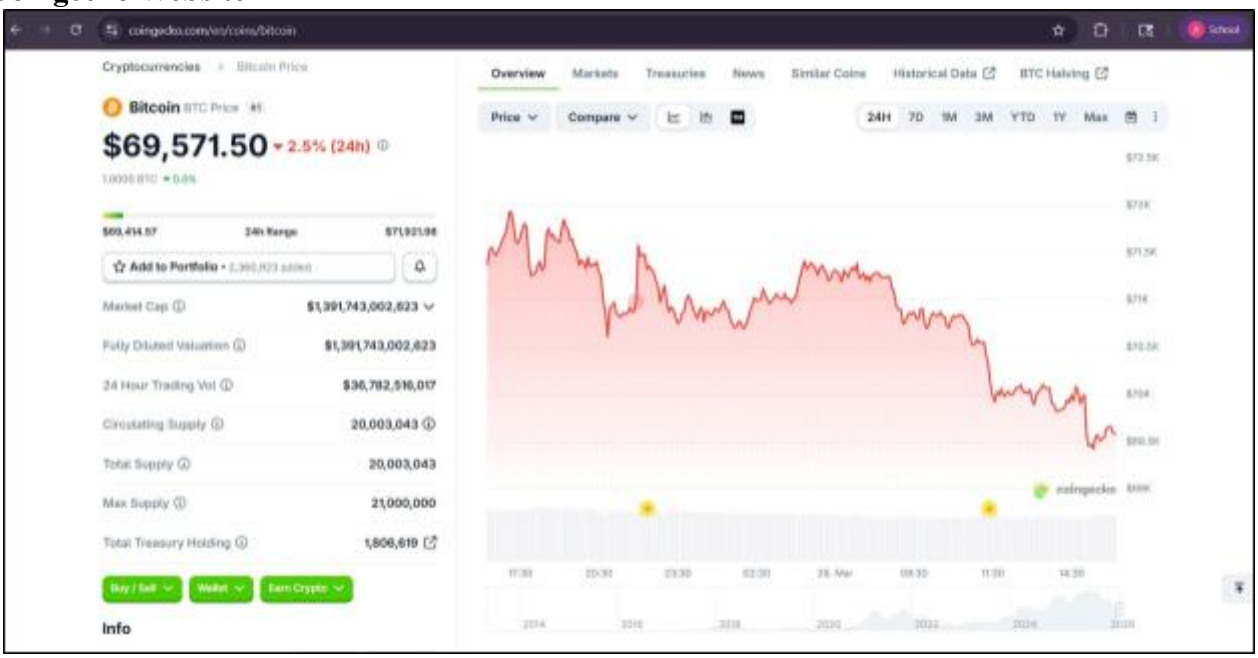
**Output:-**

CryptoCompare: LINK Analysis Dashboard
Historical Price Trend (USD)
Price vs Trading Volume
Daily Returns Distribution
Trend Analysis (Moving Averages)

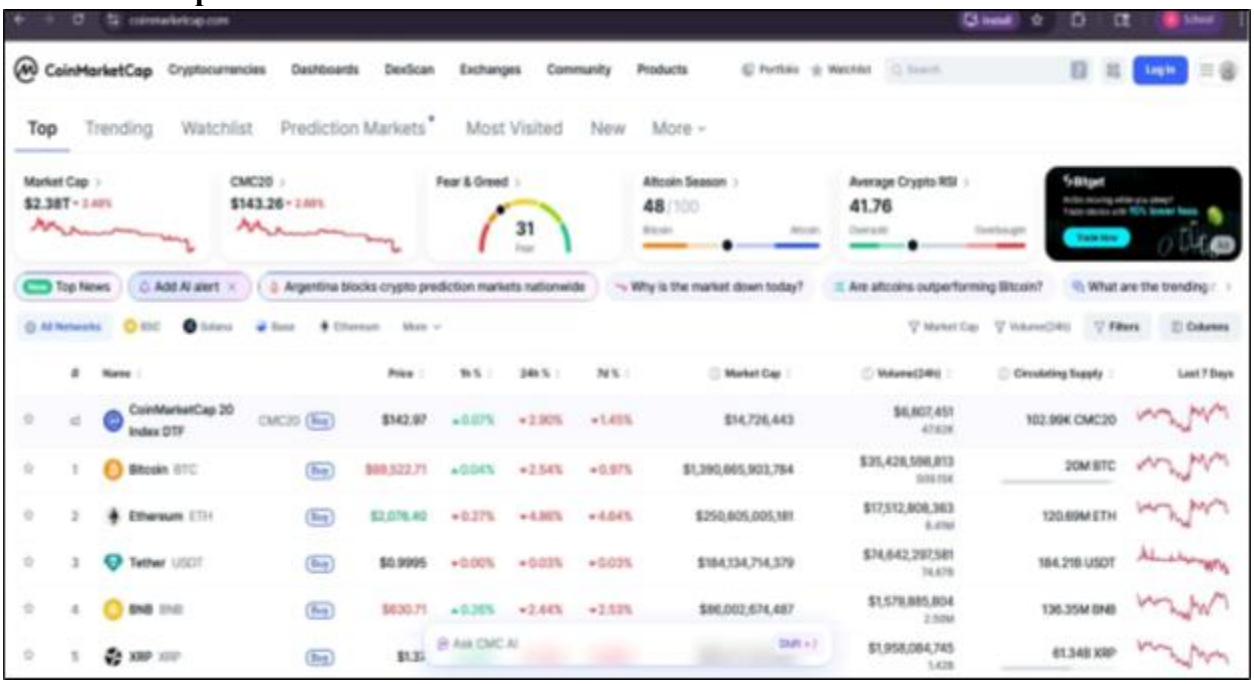

Cumulative Growth
7-Day Rolling Volatility (%)

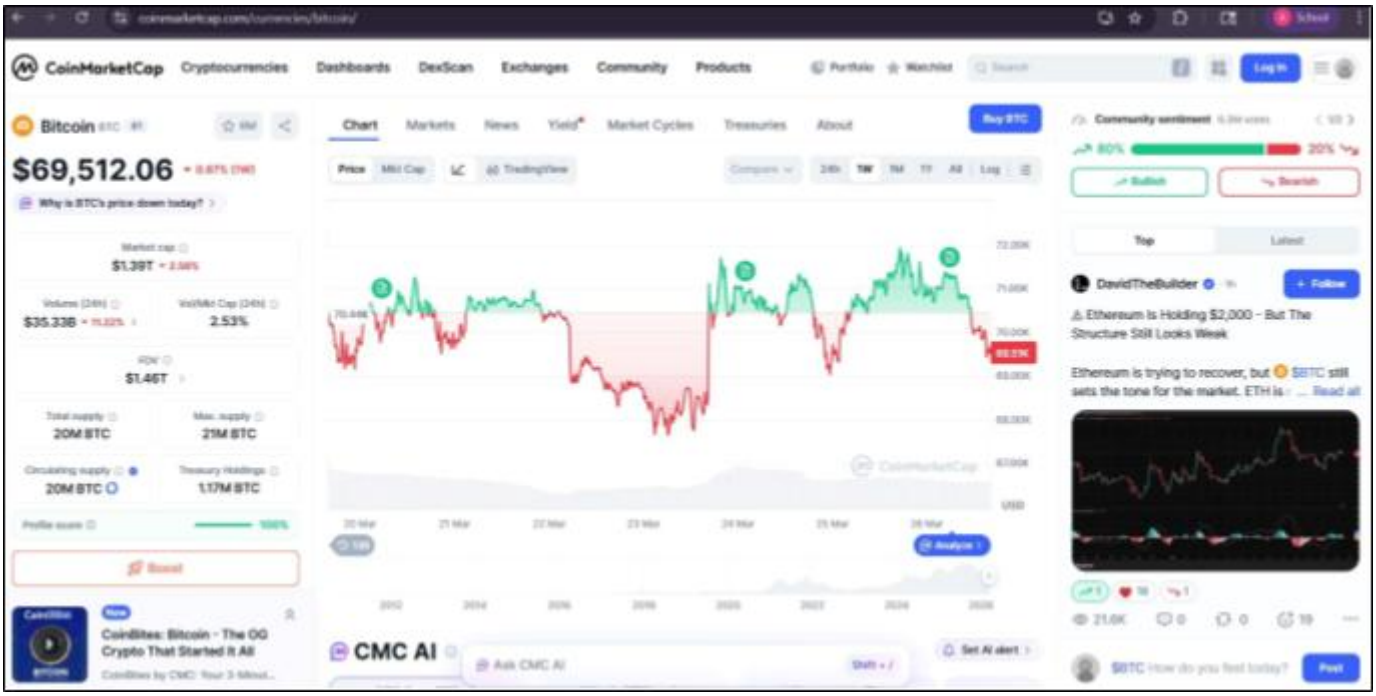

coingecko.com
Cryptocurrency Prices by Market Cap
Highlights
The global cryptocurrency market cap today is $2.46 Trillion, a 2.6% change in the last 24 hours. Read more
$2,460,001,807,038
Market Cap 2.6%
$90,030,739,656
24h Trading Volume
Trending
View more
BNB $631.10 2.4%
Sun Token $0.01741 2.4%
Bittensor $334.58 6.0%
Top Gainers
View more
Perle $0.2008 77.2%
Katana $0.01489 39.2%
Block Street $0.1576 31.1%
All
Highlights
Base Ecosystem
Categories
Options
Prediction Markets
Proof of Work (PoW)
Customize
Coin
Price
1h
24h
7d
24h Volume
Market Cap
Last 7 Days
1 Bitcoin BTC Buy $69,577.57 0.2% 2.5% 0.5% $36,764,768,445 $1,391,793,106,761
2 Ethereum ETH Buy $2,077.69 0.1% 4.8% 3.8% $16,648,195,504 $250,904,909,586
3 Tether USDT $0.9995 0.0% 0.0% 0.1% $62,409,405,710 $184,121,209,676
4 BNB BNB Buy $630.90 0.3% 2.4% 2.0% $928,001,445 $86,041,505,487
5 XRP XRP Buy $1.38 0.1% 3.1% 5.6% $2,081,894,354 $84,346,217,531

## Coingecko Website




## CoinMarketCap website






**Conclusion -**

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.